

AMRANE ANES

AI Engineer | Machine Learning | Full-Stack Developer
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SUMMARY

AI Engineering graduate specialising in machine learning, computer vision, nlp and graph neural networks. Built and deployed an end-to-end multimodal AI pipeline for solar panel defect detection and maintenance recommendation as a graduation project, covering deep learning, model fine-tuning, knowledge graph construction, and RGAT-based classification. Complementary full-stack development background with production web and NLP projects. Seeking an AI/ML or software engineering role with an international or remote-friendly team.

EDUCATION

Master's in Intelligent Systems Engineering (AI) | Université Saad Dahleb Blida

September 2024 – June 2026 | Algeria

- Graduation project: AI-Based Predictive Maintenance for Solar Panels Using Multimodal Knowledge Graph.
- Implemented EfficientNet-B0 fine-tuning, preprocessing (CLAHE), MMKG construction linking visual features and structured metadata, and Relational Graph Attention Network (RGAT) training for fault classification.
- Core coursework: Deep Learning, Graph Neural Networks, Computer Vision, Intelligent Agents & Multi-Agent Systems.

Bachelor's in Computer Science | Université Saad Dahleb Blida

Graduated 2024 | Algeria

- Final-year project: Tabibi — full-stack web platform for online medical appointment booking.

TECHNICAL SKILLS

AI / ML: PyTorch, PyTorch Geometric, EfficientNet, RGAT, Scikit-learn, spaCy, OpenCV, NumPy, Pandas

Knowledge Graphs & GNNs: MMKG construction, graph attention networks (GAT / RGAT), property graphs

NLP: Text classification, sentiment analysis, abusive speech detection & rewriting, Arabic NLP

Programming Languages: Python, JavaScript, TypeScript

Web & Backend: Next.js, React, Node.js, Express.js, MySQL, Tailwind CSS, REST APIs

Tools & DevOps: Git, GitHub, Docker (basics), Vercel, n8n (workflow automation), VS Code

PROJECTS

AI-Based Predictive Maintenance for Solar Panels Using Multimodal Knowledge Graph | Master's Graduation

Project defended with Excellence

Juin 2026

- Built an end-to-end multimodal pipeline (EL image preprocessing, EfficientNet-B0 fine-tuning, N-MMKG construction in PyTorch Geometric, RGAT training) for 5-class solar cell defect classification across 15,900 cells / 265 panels.
- Achieved F1-Macro of 0.8092, outperforming RGCN and RGAT+MLP baselines; ablation study showed visual (EL image) modality is the primary discriminative signal (F1 drops to 0.21 without it).
- Deployed as an interactive Streamlit inference dashboard; incubated as a startup (SolarGuard) at CDE Blida 1 under Ministerial Decree 1275.

AI News Automation Agent | Personal Project

2026

- Built a reactive agent pipeline in n8n: scrapes Hacker News AI threads, processes content via Groq LLM, and publishes French-language summaries to Discord — fully automated.
- Framed and documented the system using the Wooldridge & Jennings agent-property framework (reactivity, proactivity, social ability).

Real Time Fire And Smoke Detection | University Project

2026

- Built a YOLOv8-based real-time fire and smoke detection system.
- Achieved mAP@0.5 of 0.803 on the validation dataset.
- Applied data augmentation, dataset cleaning, explainable AI techniques, and model comparison experiments.

Abusive-to-Friendly Speech Transformer | NLP Project

2025

- Developed an NLP pipeline to detect hostile or abusive text and rewrite it in a neutral tone; supports Arabic and English.
- Includes a CLI sentiment analysis tool for both languages.

PDF-Based AI Chatbot (RAG) | Personal Project

2025

- Built a retrieval-augmented generation chatbot that ingests PDF documents and answers natural-language questions via an NLP pipeline (chunking, embedding, retrieval).

Tabibi — Medical Appointment Platform | Bachelor's Final-Year Project

Juin 2024

- Full-stack web application: Node.js / Express REST API, MySQL database, React / Tailwind CSS frontend.
- Features: doctor scheduling, patient records, appointment booking and management.

Personal Portfolio | anes-amrane.vercel.app

2024

- Next.js + TypeScript + Tailwind CSS, dark mode, GitHub API integration, blog, and downloadable resume.

EXPERIENCE

Developed multiple end-to-end AI and software engineering projects, including computer vision, NLP, knowledge graphs, and full-stack applications.

LANGUAGES

Arabic: Native

English: B2 — Professional working proficiency

French: B1 — Intermediate

ADDITIONAL

- Open to relocation and fully remote positions; actively exploring international AI engineering opportunities.
- Consistent self-learner: built multiple production-quality projects independently alongside full-time academic work.
- Interests: generative AI, agent-based systems, graph neural networks, startup culture, machine learning and computer vision and nlp.